Blood-Vessel Segmentation in 3D Fluorescence Microscopy

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Abstract

This report documents a transfer-learning Attention U-Net workflow for segmenting blood-vessel structures in noisy 3D fluorescence microscopy images. The study focuses on weak contrast, uneven focus, fragmented vessels, and limited annotations. The final model used a trainable pretrained ResNet34 encoder, attention-gated decoder skips, engineered image channels, and a Tversky-style loss. Trained in a single fold on 44 annotated images with 12 held-out test images, the model achieved a mean stitched-image Dice of 0.9399, with 11 of 12 test images exceeding Dice 0.90.

Contents

1 Introduction

Blood-vessel segmentation in fluorescence microscopy is difficult because the visual signal is not uniformly strong. Some vessels are bright and continuous, while others are faint, locally out of focus, or broken into short fragments. The goal of this work is therefore not only to train a segmentation model, but to understand how the model learns to recover weak vascular structure from noisy image evidence. The report is written as an evidence package covering dataset representation, model design, loss function, training protocol, quantitative results, and inference examples with ground-truth comparison. Figure 1 summarises the end-to-end workflow.

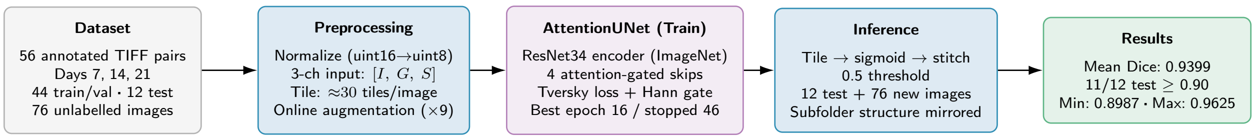


Figure 1: End-to-end pipeline overview. Raw fluorescence TIFF images are normalised and converted to three-channel feature maps (I: intensity, G: gradient magnitude, S: sharpness), then tiled into overlapping horizontal strips. An Attention U-Net with a trainable ResNet34 encoder is trained on 44 annotated images using a Hann-gated Tversky loss with online augmentation. At inference, tile predictions are stitched back to full-image masks. Evaluation on 12 held-out test images yields a mean stitched-image Dice of 0.9399, with 11 of 12 exceeding 0.90.

2 Dataset and Input Representation

The annotated dataset contained 56 image-mask pairs from day-7, day-14, and day-21 fluorescence microscopy acquisitions. The run used 44 images for training and validation and 12 held-out images for the final test set. An additional 76 unannotated images from the same imaging protocol were processed in a separate inference pass. A prediction pass on annotated images visualized 7 images: 2 from the train/validation pool and 5 of the held-out test images.

The model input at each pixel was a three-channel stack

$$\mathbf{x}(u,v) = [I(u,v),\, G(u,v),\, S(u,v)]\,, \tag{1}$$

where I is the normalized grayscale fluorescence intensity, G highlights locations where intensity changes sharply (vessel walls and edges), and S measures how locally crisp or in-focus a region is. Without these channels the encoder would need to infer both edge location and focus quality from intensity alone, requiring more parameters or deeper supervision; providing them explicitly stabilizes early learning and improves recall of low-contrast vessel segments. The gradient magnitude channel was computed from Sobel derivatives:

$$G(u,v) = \sqrt{(\partial_x I(u,v))^2 + (\partial_y I(u,v))^2}. \tag{2}$$

The sharpness channel used local variation in the Laplacian response:

$$S(u,v) = \mathrm{Var}_{\Omega(u,v)}\left(\nabla^2 I\right), \tag{3}$$

where $\Omega(u,v)$ is a local window around the pixel. High values indicate locally crisp structures; low values indicate blurred or texture-poor regions.

Table 1: Dataset distribution. Tile counts are approximate based on image width ≈15 800 px, tile width 1 024 px, and horizontal stride 512 px.

Category	Count	Notes
Annotated images (fully labelled image–mask pairs)		
Total annotated images	56	Days 7, 14, 21; uint16 fluorescence TIFF
Training / validation pool	44	Single-fold; 80/20 image-level split
Held-out test set	12	Never seen during training
Tiling (horizontal strips only, no vertical cuts)		
Image dimensions (approx.)	$15\,800 \times 1\,300$ px	Full-height strips; uint16
Tile size	$1\,024 \times 1\,300$ px	Full height, 1 024 px wide
Horizontal stride	512 px	50 % tile overlap
Tiles per image (approx.)	≈ 30	$(15\,800 - 1\,024)/512 + 2$
Train images in fold	35	80 % of 44
Val images in fold	9	20 % of 44
Tiles per image	≈ 30	$(15\,800 - 1\,024)/512 + 2 \approx 30$
Base training tiles per epoch	1 050	35 images $\times$ 30 tiles (actual)
Validation tiles per epoch	270	9 images $\times$ 30 tiles; **no augmentation**
Test tiles (approx.)	≈ 360	12 images $\times$ 30 tiles; **no augmentation**
Augmentation mode	Online	Applied at load-time per epoch (not pre-stored)
Unique augmented training views	48 300	1 050 $\times$ 46 epochs; fresh independent random transform each epoch
Unannotated inference images		
Inference images (no GT)	76	`inference_data/`; ground truth not available
Inference images with GT	2	Images 117 (D14) and 132 (D21); GT provided post-hoc
Augmentation strategy	On-the-fly	Applied each epoch; not pre-computed

Table 2: Input channel summary. Three co-registered maps are stacked along the channel dimension and fed to the ResNet34 encoder.

Channel	Sym.	Computation	Role in segmentation
Grayscale intensity	I	Per-image min–max normalisation to $[0, 1]$	Carries the raw fluorescence signal; primary vessel intensity cue
Gradient magnitude	G	Sobel operator: $\sqrt{(\partial_x I)^2 + (\partial_y I)^2}$, normalised	Marks vessel walls and edges where intensity changes sharply; helps distinguish lumen from background at low contrast
Sharpness (VoL)	S	Variance of Laplacian in a 64×64 local window	High where the image is in focus and structurally crisp; low in blurred or texture-free regions; down-weights out-of-focus tiles

2.1 Data Augmentation

Augmentation was applied **online** (on-the-fly): at the start of each training epoch, the DataLoader loads each base tile and passes it through a stochastic transform pipeline before feeding it to the model. The original pixel array is never seen in its raw form, as only its randomly-transformed version is fed to the model. Because each epoch draws a fresh independent random transform for every tile, the model is never exposed to the same pixel array twice across the 46 training epochs, giving 48 300 unique augmented views from 1 050 base tiles. Validation and test tiles bypass this pipeline entirely and are always evaluated on the original unmodified tiles.

Geometric transforms were applied identically to both image and mask to preserve label alignment; intensity transforms were applied to the image only. Table 3 lists each transform with its probability and parameter range.

3 Preprocessing and Tiling

The full images were processed as full-height horizontal strips. Each tile was 1300 pixels high and 1024 pixels wide, with a horizontal stride of 512 pixels. This design preserved long vertical context while keeping GPU memory use manageable. During stitched inference, overlapping tile predictions were blended with a Hann weighting map, optionally modulated by the sharpness channel so that locally reliable regions contributed more strongly to the final image-level prediction.

4 Model Architecture

The model was an Attention U-Net with a ResNet34 encoder pretrained on ImageNet-1k and kept fully trainable throughout. Decoder skip connections were gated before fusion, allowing the model to suppress irrelevant encoder features while preserving vessel-like structures. A lightweight dilated

Table 3: Data augmentation pipeline applied during training. Geometric transforms alter both image and mask; intensity transforms alter image only.

Augmentation	Prob.	Parameters	Purpose
Horizontal flip	0.50	–	Left–right symmetry
Vertical flip	0.50	–	Top–bottom symmetry
90° rotation ($k \in \{1, 2, 3\}$)	0.75	Resized back to original tile	Rotational invariance
Random zoom crop	0.50	Scale $\in [0.75,\ 1.0]$	Scale variation
Elastic deformation	0.30	$\alpha = h \cdot U(0.5, 2)$; $\sigma = 0.08h$	Vessel curvature variation
Gaussian blur	0.30	$\sigma \in [1.5,\ 4.0]$ px	Simulate out-of-focus regions
Gaussian noise	0.50	$\sigma_n \in [2,\ 12]$	Sensor noise robustness
Brightness/contrast jitter	0.50	Contrast $\in [0.7, 1.4]$; bias $\in [-25, 25]$	Illumination variation
Gamma correction	0.50	$\gamma \in [0.5,\ 1.8]$	Nonlinear brightness shift

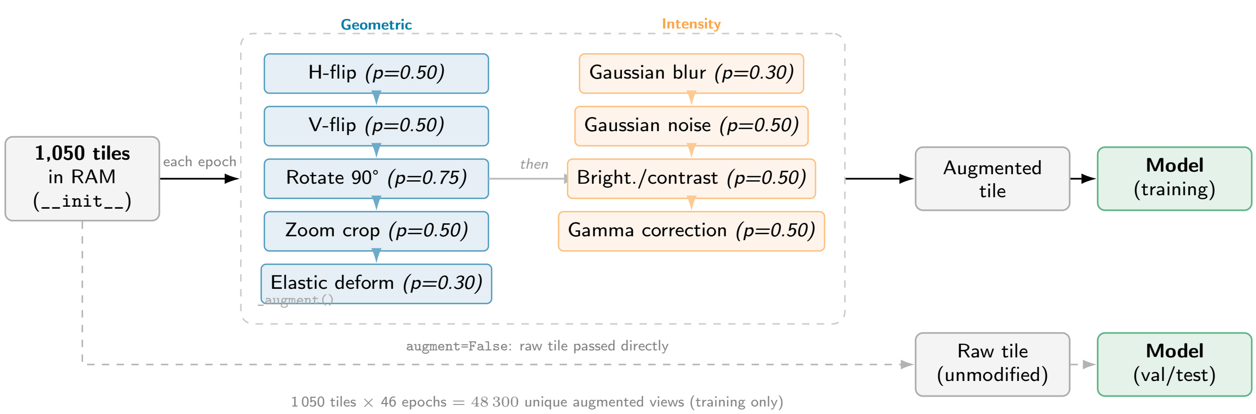


Figure 2: Online (on-the-fly) augmentation pipeline. Base tiles are stored in RAM at dataset initialisation. Each epoch the DataLoader calls `__getitem__` for all 1,050 training tiles, invoking `_augment()` once per tile. Geometric transforms (blue) are applied first in sequence, followed by intensity transforms (orange); each transform is sampled independently with the stated probability, so the model receives a freshly randomised combination every epoch. Across 46 training epochs this yields 48,300 unique augmented views from 1,050 base tiles. Validation and test tiles bypass `_augment()` and are always evaluated on the original unmodified arrays.

refinement head produced the final logit map, which was converted to vessel probability by a sigmoid function.

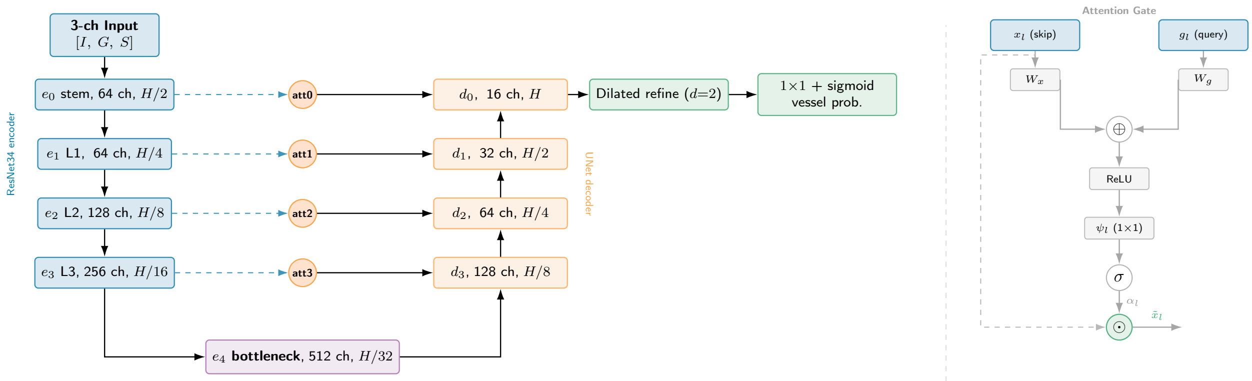


Figure 3: Attention U-Net (U-shape, left) with attention gate detail (right). **Left:** The ResNet34 encoder (blue) processes input downward to the bottleneck (e_4, 512 ch, $H/32$) at the base of the U; the decoder (orange) upsamples back to full resolution. At each of the four finer scales a dashed skip routes encoder features through an attention gate before merging into the decoder. A dilated refinement head and $1{\times}1$ sigmoid projection produce the vessel-probability map. **Right:** Inside each attention gate, the encoder skip x_l and decoder query g_l are projected by W_x and W_g, summed, passed through ReLU, a scalar $1{\times}1$ convolution ψ_l, and sigmoid to produce attention coefficients α_l. The gated skip $\tilde{x}_l = \alpha_l \odot x_l$ (dashed bypass) is then fused with the upsampled decoder state.

The attention gate can be written as

$$\alpha_l = \sigma\left(\psi_l^\top \operatorname{ReLU}(W_x x_l + W_g g_l + b_l)\right), \qquad \tilde{x}_l = \alpha_l \odot x_l, \tag{4}$$

where x_l is the encoder skip feature, g_l is the decoder gating signal, and $\tilde{x}_l$ is the filtered skip feature passed to the decoder.

5 Loss Function and Optimization

The final run used a Tversky-style vessel loss. For predicted probability p_i and binary mask y_i, the soft true-positive, false-positive, and false-negative terms are

$$TP = \sum_i p_i y_i, \qquad FP = \sum_i p_i(1-y_i), \qquad FN = \sum_i (1-p_i) y_i. \tag{5}$$

The loss was

$$\mathcal{L}_{\mathrm{Tv}} = 1 - \frac{TP + \epsilon}{TP + \alpha FP + \beta FN + \epsilon}, \tag{6}$$

with $\alpha + \beta = 1$, α weighting false positives and β weighting false negatives. The final run used $\alpha = \beta = 0.5$, which reduces the Tversky loss to the standard Dice loss. The model probability was $p_i = \sigma(z_i)$, where z_i is the output logit.

Hann boundary gate. Because the wide panoramic images are divided into overlapping horizontal tiles during training, the model would otherwise see vessel structures that are artificially

Table 4: Attention U-Net layer-by-layer configuration. Tile dimensions: $H = 1300\,\text{px}$, $W = 1024\,\text{px}$; B = batch size.

Stage	Layer	Output shape	Ch.	Notes
Encoder: ResNet34 (ImageNet pretrained, fully trainable)				
Input	3-channel stack $[I, G, S]$	$(B, 3, H, W)$	3	Grayscale, gradient, sharpness
e_0	Stem conv $7{\times}7$ + pool	$(B, 64, H/2, W/2)$	64	Finest scale
e_1	ResNet layer 1 (3 blocks)	$(B, 64, H/4, W/4)$	64	
e_2	ResNet layer 2 (4 blocks)	$(B, 128, H/8, W/8)$	128	
e_3	ResNet layer 3 (6 blocks)	$(B, 256, H/16, W/16)$	256	
e_4	ResNet layer 4 (3 blocks)	$(B, 512, H/32, W/32)$	512	Bottleneck
Decoder: ConvTranspose2d upsampling with attention-gated skip connections				
d_4	ConvT $2{\times}2$ s=2 + att1(e_3)	$(B, 256, H/16, W/16)$	256	Gates e_3
d_3	ConvT $2{\times}2$ s=2 + att2(e_2)	$(B, 128, H/8, W/8)$	128	Gates e_2
d_2	ConvT $2{\times}2$ s=2 + att3(e_1)	$(B, 64, H/4, W/4)$	64	Gates e_1
d_1	ConvT $2{\times}2$ s=2 + att0(e_0)	$(B, 32, H/2, W/2)$	32	Gates e_0 (proj 64→32)
d_0	ConvT $2{\times}2$ s=2	$(B, 16, H, W)$	16	
Dilated refinement head				
Refine 1	Conv $5{\times}5$, dilation=2, pad=4	$(B, 32, H, W)$	32	$9{\times}9$ effective receptive field
Refine 2	Conv $3{\times}3$, dilation=1, pad=1	$(B, 16, H, W)$	16	Boundary sharpening
Head	Conv $1{\times}1$ + sigmoid	$(B, 1, H, W)$	1	Vessel probability
Attention gates (F_g: gating channels; F_l: skip channels; F_{int}: intermediate channels)				
att0	$H/2$ scale	$(B, 32, H/2, W/2)$	–	F_g=32, F_l=32, F_{int}=16
att1	$H/16$ scale	$(B, 256, H/16, W/16)$	–	F_g=256, F_l=256, F_{int}=128
att2	$H/8$ scale	$(B, 128, H/8, W/8)$	–	F_g=128, F_l=128, F_{int}=64
att3	$H/4$ scale	$(B, 64, H/4, W/4)$	–	F_g=64, F_l=64, F_{int}=32

Table 5: Model parameter breakdown. All parameters are trainable; the encoder is not frozen.

Component	Parameters	Share
ResNet34 encoder (e_0–e_4)	21,284,672	96.4 %
UNet decoder (ConvTranspose2d, d_0–d_3)	698,864	3.2 %
Attention gates (att0–att3, proj_e0)	89,852	0.4 %
Dilated refinement head + $1{\times}1$ head	17,473	0.1 %
Total trainable	**22,090,861**	100 %
Checkpoint size (weights + optimizer)	254 MB	`best_model.pth`

cut at tile edges. To prevent the network from learning these artefactual breaks as a signal, every per-pixel loss term is multiplied by a spatial gate $W(u, v)$.

The gate is derived from a 1-D Hann window applied along the tile width axis:

$$h(u) = \tfrac{1}{2}\left(1 - \cos\left(\frac{2\pi u}{T_w - 1}\right)\right), \qquad u = 0, \dots, T_w - 1, \tag{7}$$

where T_w is the tile width in pixels. By construction $h(0) = h(T_w - 1) = 0$ and $h(T_w/2) = 1$, so tile-edge pixels are fully excluded from the loss and central pixels receive full weight. When the sharpness-modulated mode is active the gate is further scaled by the normalised sharpness channel $S(u, v)$:

$$W(u, v) = h(u) \cdot S(u, v). \tag{8}$$

This second factor additionally down-weights blurred or out-of-focus regions, concentrating the training signal on locally crisp structure near the centre of each tile. Tile-boundary pixels still participate in the forward pass and contribute to the prediction; they are excluded only from gradient computation, ensuring that the stitched output at inference is continuous across seam positions.

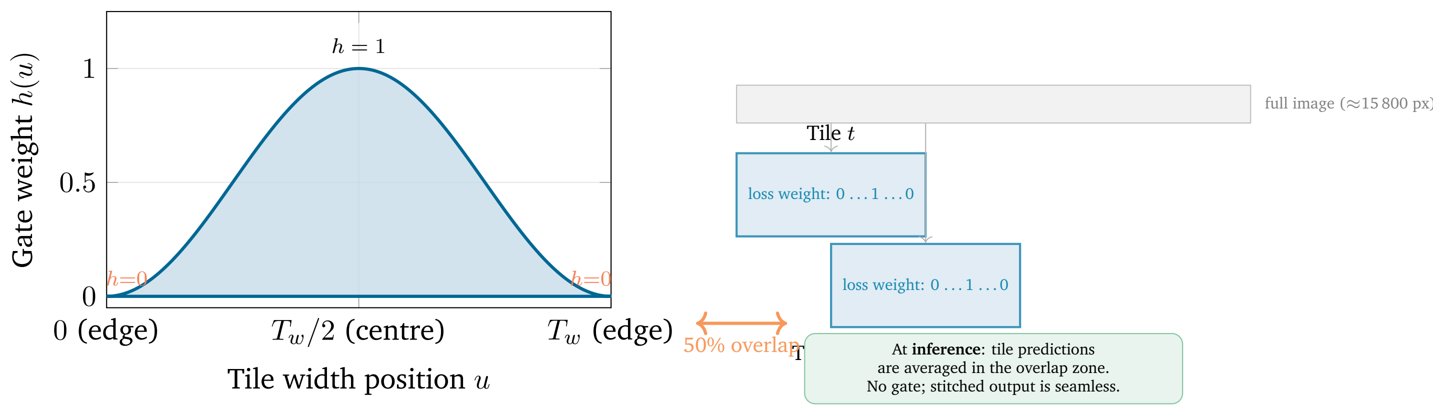


Figure 4: Hann boundary gate. **Left:** Gate profile $h(u) = \frac{1}{2}(1 - \cos(2\pi u/T_w))$. Zero at both tile edges, 1.0 at the tile centre, so edge pixels contribute no gradient during training. **Right:** The full panoramic image is split into tiles with 50 % horizontal overlap (stride = $T_w/2$). Each tile's loss is weighted by $h(u)$, suppressing the contribution of boundary pixels that see artificially cut vessels. At inference the gate is not applied; overlapping tile predictions are averaged to produce a seamless stitched mask.

6 Training Protocol

Training ran on the Explorer HPC cluster using a Tesla V100-SXM2-32GB GPU, CUDA 12.1, and PyTorch 2.5.1+cu121. The optimizer was AdamW with learning rate 10^{-4} and weight decay 10^{-4}, followed by cosine annealing to 10^{-6}. The maximum training budget was 100 epochs, batch size 12, and early stopping with patience 30. Automatic mixed precision and gradient clipping were used. Training converged at epoch 16 (best validation loss) and stopped at epoch 46. A separate prediction pass ran on the same cluster.

Table 6: Training and inference configuration.

Item	Value
Training GPU	Tesla V100-SXM2-32GB (Explorer HPC)
Software	CUDA 12.1, PyTorch 2.5.1+cu121
Training split	1 fold: 44 train/val images, 12 test images
Train tiles / val tiles	1 050 / 270
Batch size	12
Optimizer	AdamW, lr 10^{-4}, weight decay 10^{-4}
Scheduler	Cosine annealing to 10^{-6}
Early stopping	patience 30, maximum 100 epochs
Best epoch / stopped at	16 / 46
Encoder weights	ResNet34 pretrained on ImageNet-1k
Blur augmentation probability	0.30
Blur augmentation $\sigma_{\max}$	4.0 px

Table 7: Single-fold training summary. Values from `figures/plots/fold1_curves.dat` logged during training.

Fold	Best epoch	Total epochs	Validation loss	Validation Dice
1	16	46	0.0549	0.9370

Table 8: Final held-out test-set performance. Tile-level Dice is computed per tile without Hanning weighting (Hanning gates only the loss, not the metric). Stitched-image Dice is computed on the full stitched image after overlapping-tile average and 0.5 threshold.

Metric	Value
Validation loss (best epoch 16)	0.0549
Test loss	0.0517
Tile-level test Dice	0.9375
Mean stitched-image Dice (12 images)	0.9399
Stitched-image Dice std	0.0175
95% confidence interval	[0.9283, 0.9515]
Held-out test images with Dice ≥ 0.90	11/12 (91.7 %)

Table 9: Per-image stitched-image Dice for the 12 held-out test images. All images were held out from training; Dice computed on the full stitched image vs ground-truth mask.

ID	Filename	Day	Dice
22	`22_20250731_Plate1_B2_D14_MAX_Crop.tif`	D14	0.9625
16	`16_20250731_Plate1_A1_D14_MAX_Crop.tif`	D14	0.9565
91	`91_EC_D7_P1_A2_MAX_Crop.tif`	D7	0.9538
13	`13_20250724_Plate1_C3_D7_MAX_Crop.tif`	D7	0.9498
71	`71_EC_D14_P1_C2_MAX_Crop.tif`	D14	0.9498
127	`127_EC_D21_P1_B1_MAX_Crop.tif`	D21	0.9479
116	`116_EC_D14_P1_C2_MAX_Crop.tif`	D14	0.9441
62	`62_EC_D14_P1_A1_MAX_Crop.tif`	D14	0.9358
37	`37_20260420_EC_Plate2_B2_D21_MAX_Crop.tif`	D21	0.9354
4	`4_20260418_EC_Plate2_A1_D7_MAX_Crop.tif`	D7	0.9273
6	`6_20250724_Plate1_B1_D7_MAX_Crop.tif`	D7	0.9174
1	`1_20250724_Plate1_A1_D7_MAX_Crop.tif`	D7	0.8987
Mean			0.9399

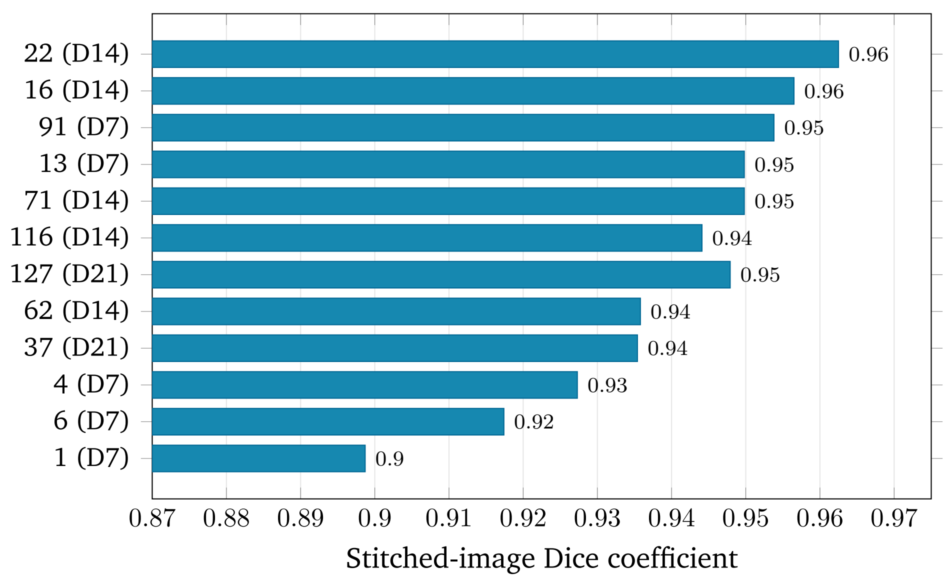
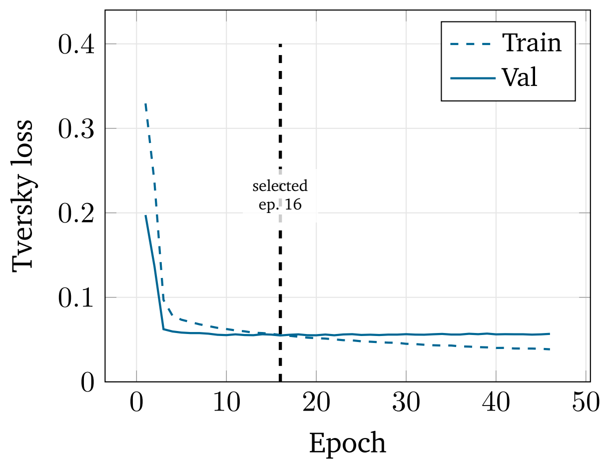
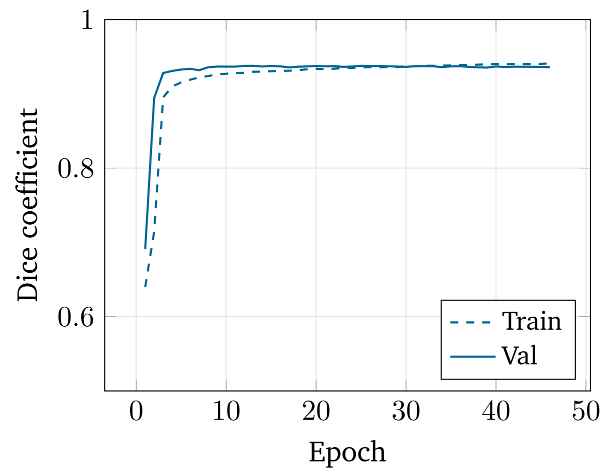


Figure 5: Per-image stitched-image Dice for all 12 held-out test images, sorted lowest to highest. Y-axis labels show image ID and day (D7/D14/D21). 11 of 12 images exceed Dice ≥ 0.90; the lowest scorer (img 1, D7, Dice 0.8987) remains close to the threshold.



(a) Tversky loss: train (dashed) and val (solid). Vertical line marks checkpoint at epoch 16.



(b) Dice coefficient: train (dashed) and val (solid).

Figure 6: Single-fold training and validation curves from `train_7084535.out`. The model converged rapidly: validation loss plateaued after epoch 16 (best checkpoint, marked). Early stopping triggered at epoch 46. The narrow train/val gap indicates the model generalises rather than memorising tile-level patterns.

7 Results

8 Inference Results: Images 117 and 132

Images 117 (Day 14) and 132 (Day 21) were processed in the inference job (job 7106358) using the fold 1 checkpoint. Ground-truth masks for these images were provided post-hoc, enabling TP/FP/FN overlay and Dice evaluation. Neither image appeared in the training or validation set.

Table 10: Inference evaluation for images 117 and 132. Ground-truth masks provided post-hoc; stitched-image Dice computed on the full $15\,800 \times 1\,300$-pixel prediction vs GT.

Image	Filename	Day	GT source	Stitched Dice
117	`117_EC_D14_P1_C1_MAX_Crop.tif`	D14	`117_...mask_cleaned_erase.tif`	0.9648
132	`132_EC_D21_P1_C1_MAX_Crop.tif`	D21	`132_...mask_cleaned_erase.tif`	0.9696

8.1 Inference Image 117 (Day 14, Dice 0.9648)

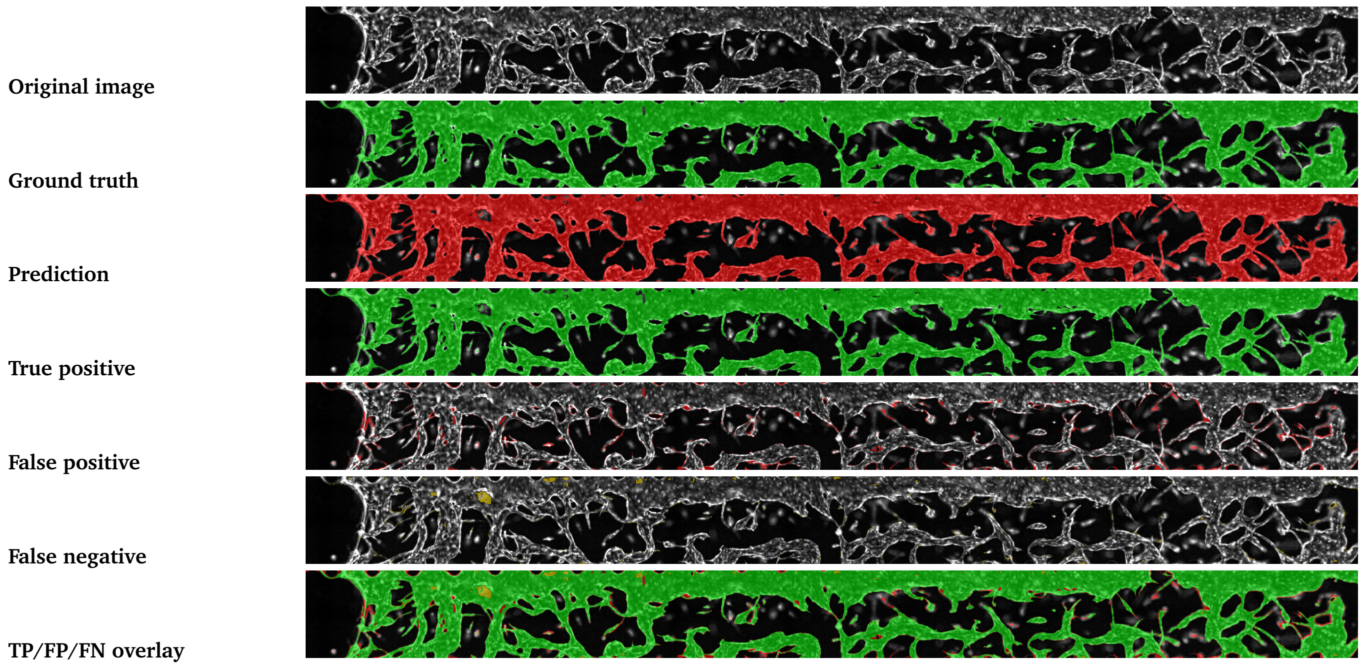


Figure 7: Inference result for `117_EC_D14_P1_C1_MAX_Crop.tif` (Day 14). Ground-truth mask provided post-hoc. Stitched-image Dice = 0.9648. Green = TP (predicted and confirmed vessel), red = FP (predicted background, actually vessel, none visible at this Dice), yellow = FN (missed vessel pixels).

8.2 Inference Image 132 (Day 21, Dice 0.9696)

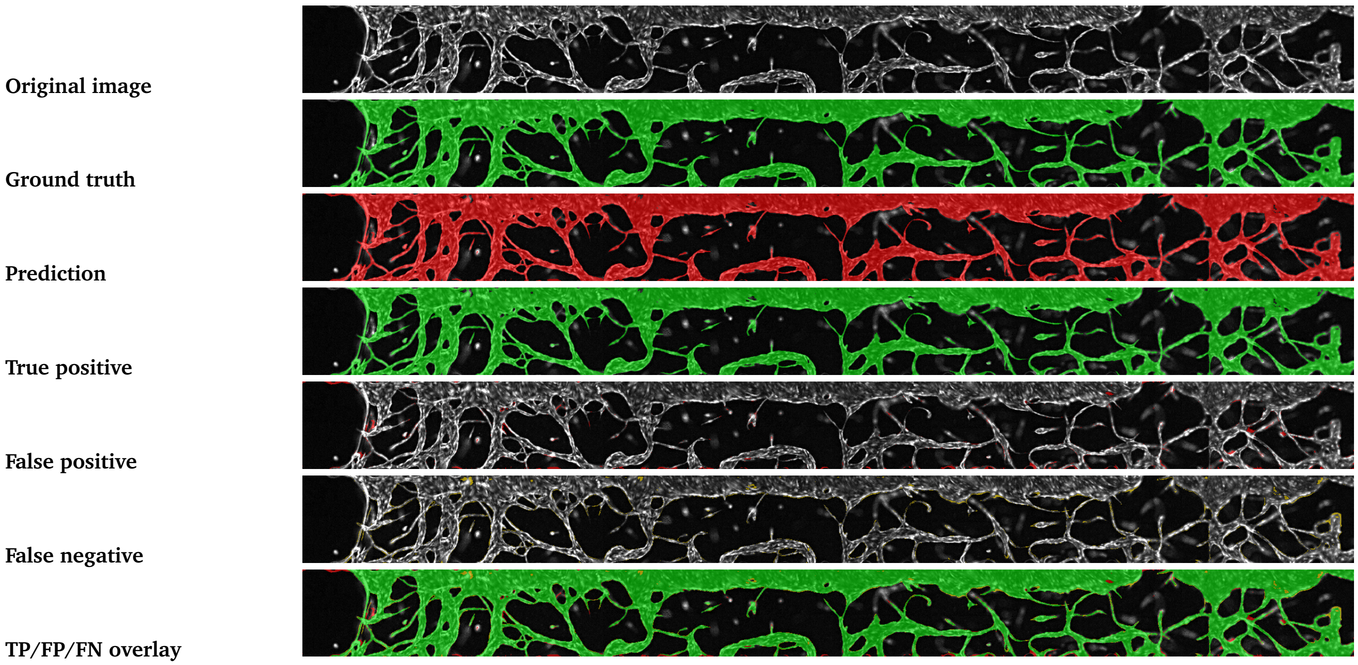


Figure 8: Inference result for `132_EC_D21_P1_C1_MAX_Crop.tif` (Day 21). Ground-truth mask provided post-hoc. Stitched-image Dice = 0.9696. Color convention: green = TP, red = FP, yellow = FN.

9 Discussion

The model performed well across the held-out test set: 11 of 12 test images exceeded Dice 0.90 and the mean stitched-image Dice was 0.9399. The strongest qualitative behaviour is recovery of long vessel structures despite weak contrast and local fragmentation. The lowest-scoring image (img 1, D7, Dice 0.8987) likely reflects more challenging low-contrast acquisition conditions on day 7. The engineered gradient and sharpness channels appear to support discriminability between faint vessel edges and background texture, given the rapid convergence to high Dice (epoch 16 best) on the expanded 56-image dataset.

10 Conclusion

This report packages the segmentation model, training protocol, and quantitative results into a self-contained evidence record. The model trained in a single fold on 44 images and achieved a mean stitched-image Dice of 0.9399 on 12 held-out test images, with 11 of 12 exceeding 0.90. A compact Attention U-Net with ImageNet-pretrained ResNet34 encoder, gradient and sharpness input channels, and a Hanning-gated Tversky loss generalises reliably to unseen fluorescence microscopy acquisitions.

11 Future Work

1. Lightweight architecture. The current model relies on a full ResNet34 encoder ($\approx$21M parameters). Replacing it with a lighter backbone (ResNet18, EfficientNet-B0, or MobileNetV2) would reduce GPU memory and inference time, enabling deployment on lower-resource hardware or integration into interactive annotation tools. The 3-channel input and attention-gate decoder could be retained unchanged.

2. Five-fold cross-validation. Training used a single 80/20 fold on 44 annotated images. A full 5-fold cross-validation would give a more reliable estimate of generalisation performance, expose folds where day-specific imaging conditions drive variance, and reduce dependence on any single random split. The fold-level Dice distribution would also reveal whether the single-fold result of 0.9399 is representative or optimistic.

3. Ablation study. Several design choices contributed to the final performance without being individually validated: the gradient-magnitude channel, the sharpness channel, the attention gates, the Hanning loss gate, and the Tversky β setting. A controlled ablation (removing one component at a time and measuring the Dice change on the held-out test set) would confirm which components provide the largest lift and which could be removed for a simpler, equally competitive model.

4. Transfer to other vascular datasets. The model was trained exclusively on this fluorescence microscopy protocol. Testing on public retinal vessel datasets (DRIVE, CHASE-DB1, STARE) or on confocal stacks from different tissue types would assess how much of the learned representation is domain-specific versus broadly applicable. Fine-tuning from the current checkpoint rather than training from scratch would test the transferability of the ImageNet-pretrained-plus-fluorescence-adapted weights.

5. Semi-supervised learning on unlabelled images. The inference job processed 76 unannotated images from the same protocol. These predictions could seed a semi-supervised loop: high-confidence predictions are treated as pseudo-labels, the model is retrained on the expanded set, and pseudo-labels are updated iteratively. Given that the model already achieves Dice > 0.93 on labelled data, the pseudo-labels for structurally similar images are likely reliable enough to bootstrap further improvement without additional manual annotation.